SE LAB 07
Fuzzing for Input Validation Bugs

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Deliverables

→Original processor execution

```
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7> python processor.py
==== Running validate_height_and_fastpass ====
Enter rider spec (Height:FastPass, e.g. 140:YES): 140:YES
Height 140cm: Approved (FastPass Priority)

==== Running compute_dispatch_throughput ====
Enter riders/mins (e.g. 48/2.5): 48/2.5
Throughput (riders/min): 19.2

==== Running resolve_safety_restraint_code ====
Enter ride zone code (e.g. CST_01, WTR_A): CST_01
Restraint: Over-the-Shoulder Harness
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7>
```

→Initial Hypothesis failures

```
================== short test summary info ==================
FAILED test_processor.py::test_validate_height_and_fastpass_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub...
FAILED test_processor.py::test_compute_dispatch_throughput_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub...
FAILED test_processor.py::test_resolve_safety_restraint_code_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub...
==================== 3 failed in 0.53s ====================
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7>
```

→Fixed processor execution

```
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7> python processor.py
==== Running validate_height_and_fastpass ====
Enter rider spec (Height:FastPass, e.g. 140:YES): 140:YES
Height 140cm: Approved (FastPass Priority)

==== Running compute_dispatch_throughput ====
Enter riders/mins (e.g. 48/2.5): 48/2.5
Throughput (riders/min): 19.2

==== Running resolve_safety_restraint_code ====
Enter ride zone code (e.g. CST_01, WTR_A): CST_01
Restraint: Over-the-Shoulder Harness
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7>
```

→Final Hypothesis tests passing

```
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7> python -m pytest test_processor
.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_par
k_ride_access\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py::test_validate_height_and_fastpass_no_crash PASSED                                                                [ 33%]
test_processor.py::test_compute_dispatch_throughput_safe PASSED                                                                     [ 66%]
test_processor.py::test_resolve_safety_restraint_code_safe PASSED                                                                   [100%]

============================================================ 3 passed in 1.13s ============================================================
(.venv) PS C:\Users\JPavithra\Downloads\SEMESTER 55555\SOFTWARE ENGINEERIGN\L6_L7_11_amusement_park_ride_access\Lab 7>
```